RTX 5090 Low-Precision Product-Pattern Width Report

Direct PTX `M - M + epsilon` product probes

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Headline finding. The earlier A-only pattern understated FP8 E4M3 range. With independent `A*B` products, FP8 E4M3 x E4M3 reaches an extreme test width of **36 bits**. That extreme case does not survive, and product-exponent scanning finds **27 effective bits** for all tested FP8 variants. FP6 and FP4 remain format-limited in this product-pattern probe.

GPU	NVIDIA GeForce RTX 5090, compute capability 12.0
Build target	`compute_120a` / `sm_120a`
Source	`src/tc_test_numerics-5090-lowp-reduction-pattern.cu`
Result files	`5090/result-5090-fp8-reduction-pattern.txt`, `5090/result-5090-fp6-reduction-pattern.txt`, `5090/result-5090-fp4-reduction-pattern.txt`

1 Scope

This report covers the standalone product-pattern probe listed on the title page. It is separate from the accumulator-boundary and repeated-MMA probes. The test uses direct inline PTX `mma.sync.aligned` instructions and does not call cuBLASLt.

The probe reports externally visible behavior for selected product positions. It does not claim to reconstruct the physical tensor-core reduction tree.

2 Reading Guide

This report uses three related but different quantities:

Term	Meaning	Interpretation
Extreme width	Gap between the largest finite A/B product and the smallest positive A/B product.	Widest gap the format can generate in this test.
Max effective width	Widest product-exponent gap where epsilon remains visibly retained after cancellation.	Measured retention boundary for this layout and criterion.
Format-limited	Max effective width equals extreme width.	The test ran out of representable product range before finding a failing boundary.

The key result is therefore: FP8 exposes a non-format-limited 27-bit retention boundary in this selected layout; FP6 and FP4 remain capped by their representable product ranges.

3 Method

The test maps independent product terms for output register `D0`. The selected terms are:

Role	Position	Weight in D0
`+M`	`A0.byte0 * B0.byte0`	4
`-M`	`A0.byte1 * B0.byte1`	4
`epsilon`	`A0.byte2 * B0.byte2`	4

The cross terms among those positions are checked to be absent for `D0`, so the pattern isolates three product contributions:

$$D0 = 4M - 4M + 4\epsilon$$

The test enumerates every positive finite raw value in the A format and every positive finite raw value in the B format. It forms all positive products, groups them by product exponent, and scans exponent gaps:

$$effective\ width = exponent(M) - exponent(\epsilon) + 1$$

A case counts as retained when:

```
observed > 0
abs(observed - expected) <= 0.5 * abs(expected)
```

The 50% tolerance is intentional. This pattern detects whether the small term is still visible after cancellation. It does not require exact arithmetic on the retained term, so borderline FP8 cases are reported explicitly.

The sign pattern uses a negative A value for the `-M` term and keeps the same B product magnitude. This keeps `+M` and `-M` symmetric while allowing the third independent product to carry ϵ.

4 Results

Family	Variant	Extreme width	Extreme survived	Max effective width	Format-limited
FP8	E4M3 x E4M3	36	no	27	no
FP8	E5M2 x E5M2	64	no	27	no
FP8	E4M3 x E5M2	50	no	27	no
FP8	E5M2 x E4M3	50	no	27	no
FP6	E2M3 x E2M3	12	yes	12	yes
FP6	E3M2 x E3M2	18	yes	18	yes
FP4	E2M1 x E2M1	8	yes	8	yes

4.1 Product Range

Variant	Tested pairs	Survived	Max product	Min product
FP8 E4M3 x E4M3	666	612	200704	3.8147e-06
FP8 E5M2 x E5M2	2080	1332	3.2883e+09	2.3283e-10
FP8 E4M3 x E5M2	1275	976	25690112	2.9802e-08
FP8 E5M2 x E4M3	1275	976	25690112	2.9802e-08
FP6 E2M3 x E2M3	78	78	56.25	0.015625
FP6 E3M2 x E3M2	171	171	784	0.00390625
FP4 E2M1 x E2M1	36	36	36	0.25

4.2 Survival Ratios

Variant	Survival ratio	Meaning
FP8 E4M3 x E4M3	91.9%	Wide gaps begin to fail; not format-limited.
FP8 E5M2 x E5M2	64.0%	Much wider available range; many large gaps fail.
FP8 E4M3 x E5M2	76.5%	Mixed range exposes the same 27-bit boundary.
FP8 E5M2 x E4M3	76.5%	Same boundary in the opposite operand order.
FP6 E2M3 x E2M3	100%	No failing case within representable product range.
FP6 E3M2 x E3M2	100%	No failing case within representable product range.
FP4 E2M1 x E2M1	100%	No failing case within representable product range.

4.3 Extreme Cases

Variant	Width	Observed	Expected	Survived
FP8 E4M3 x E4M3	36	0	1.5259e-05	no
FP8 E5M2 x E5M2	64	0	9.3132e-10	no
FP8 E4M3 x E5M2	50	0	1.1921e-07	no
FP8 E5M2 x E4M3	50	0	1.1921e-07	no
FP6 E2M3 x E2M3	12	0.0625	0.0625	yes
FP6 E3M2 x E3M2	18	0.015625	0.015625	yes
FP4 E2M1 x E2M1	8	1	1	yes

4.4 Max-Width Retained Cases

Variant	Width	M exp	Eps exp	Observed	Expected
FP8 E4M3 x E4M3	27	17	-9	0.0078125	0.0153809
FP8 E5M2 x E5M2	27	31	5	128	240
FP8 E4M3 x E5M2	27	24	-2	1	1.96875
FP8 E5M2 x E4M3	27	24	-2	1	1.96875
FP6 E2M3 x E2M3	12	5	-6	0.0625	0.0625
FP6 E3M2 x E3M2	18	9	-8	0.015625	0.015625
FP4 E2M1 x E2M1	8	5	-2	1	1

The FP8 max-width retained cases are edge cases. The observed value is roughly half of the expected epsilon contribution, but remains positive and within the retention tolerance. Therefore, **27 bits** means the widest detectable retained epsilon under this criterion, not exact 27-bit arithmetic.

The next wider gaps are not listed as pass cases because the test requires both a positive result and a result close enough to the expected epsilon. This avoids counting unrelated residue as a retained small term.

5 Format Analysis

5.1 FP8 E4M3 x E4M3

The earlier A-only pattern understated the available range. With A and B both participating:

```
max product = 200704
min product = 3.81469727e-06
product exponent gap = 17 - (-18) + 1 = 36
```

The extreme case is lost. The widest retained case is 27 bits, with M at exponent 17 and ϵ at exponent -9. This is not format-limited.

The retained 27-bit case is near the tolerance edge: `observed = 0.0078125` and `expected = 0.0153808594`. This is a retention signal, not an exact summation signal.

5.2 FP8 E5M2 x E5M2

E5M2 has a wider exponent range than E4M3, so the extreme product width reaches 64 bits. The extreme case is lost, and the widest retained case is again 27 bits. More representable product range does not increase the observed retained width for this pattern.

The retained 27-bit case is also near the threshold: `observed = 128` and `expected = 240`. E5M2 provides a larger test range, but the widest retained product gap is not more accurate in absolute terms.

5.3 FP8 Mixed E4M3/E5M2

Both mixed variants reach an extreme width of 50 bits and a max effective width of 27 bits. The two directions match in this selected layout, but that should not be generalized to every possible operand placement or output lane without additional layout sweeps.

The mixed cases show that the 27-bit retained-width signal is not unique to a same-format FP8 instruction under this selected independent-product layout.

5.4 FP6 E2M3 x E2M3

The entire tested product range is 12 bits, and the extreme case survives exactly. This result is format-limited: it proves the tested path preserves the full E2M3 product range used here, but it does not prove the internal retention width is only 12 bits.

5.5 FP6 E3M2 x E3M2

The tested range expands to 18 bits because E3M2 has more exponent range than E2M3. The extreme case survives exactly, so this is also format-limited. The test does not find a failing FP6 E3M2 product-pattern case.

5.6 FP4 E2M1 x E2M1

The tested product range is 8 bits. The extreme case survives exactly. This is strong evidence that the tested path can retain every product gap expressible by FP4 E2M1 in this layout, but it cannot bound a wider internal tree because the format cannot generate a wider product gap.

6 Edge Cases and Limitations

- The result is layout-sensitive. Other positions, lanes, or output registers may have different association behavior.
- FP8's 27-bit retained cases pass because epsilon remains visible and within the 50% retention tolerance, not because the result equals the expected value.
- FP6 and FP4 are format-limited in this test. Their max effective width is the largest width generated by the product format here.
- The grouping key is product exponent. Products in the same exponent bucket can have different significands.
- This test covers direct `mma.sync.aligned.m16n8k32` PTX and one selected output register. It does not cover block-scaled MXFP instructions or full GEMM accuracy.

7 Relationship To The Boundary Probe

This report and `lowp-reduction-width-report.md` measure different questions:

Probe	Pattern	Reported signal
Boundary scan	`D = C + 32`	When a full MMA update stops changing a large external accumulator.
Product pattern	`M - M + epsilon`	Whether a tiny product term survives internal cancellation.

The accumulator-boundary probe reports a 21-bit per-MMA update boundary. This product-pattern probe reports FP8 retained cases up to 27 bits in a specific cancellation layout. These values are complementary, not contradictory.

8 Suggested Follow-Up Tests

- Sweep additional independent product positions and output registers.
- Tighten the retained-case threshold to separate epsilon visibility from approximate epsilon correctness.
- Record the smallest failing gap above each max retained width.
- Add block-scaled MXFP4/MXFP6 variants when direct PTX support is available.

9 Verification

```
make -B test-5090-fp8-reduction-pattern \
        test-5090-fp6-reduction-pattern \
        test-5090-fp4-reduction-pattern

python3 run_tests.py -o 5090 \
        5090-fp8-reduction-pattern \
        5090-fp6-reduction-pattern \
        5090-fp4-reduction-pattern

compute-sanitizer --tool memcheck ./test-5090-fp8-reduction-pattern
compute-sanitizer --tool memcheck ./test-5090-fp6-reduction-pattern
compute-sanitizer --tool memcheck ./test-5090-fp4-reduction-pattern
```

The generated SASS contains direct `QMMA.16832` instructions:

Target	Confirmed instruction family
`test-5090-fp8-reduction-pattern`	`QMMA.16832.F32.E4M3/E5M2`
`test-5090-fp6-reduction-pattern`	`QMMA.16832.F32.E2M3/E3M2`
`test-5090-fp4-reduction-pattern`	`QMMA.16832.F32.E2M1`